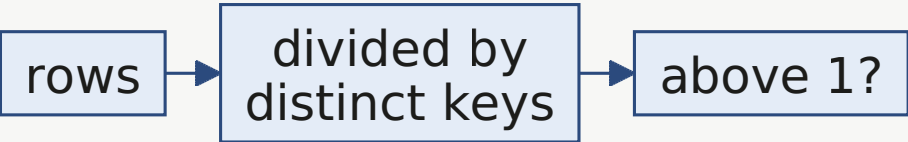


1 Grain, before anything else



| The file says  | One row is             | A pivot over it |
|----------------|------------------------|-----------------|
| customer table | One customer           | Correct         |
| raw export     | One order-payment pair | Roughly double  |

Write the grain into the filename or the first sheet. It costs one line.

**Crux.** What one row means decides whether a pivot over the file is honest, and a file browser cannot show you that.

2 A pivot is only as honest as its table

The fan-out from Tuesday did not go away. It moved into a file that makes it invisible, and filtering the pivot to one segment makes the wrong number look smaller and more plausible.

**Crux.** The pivot sums the column it is given across the rows it is given, so every check you would run on the pivot passes while the total is twice the truth.

4 Portability, which bites once

Send the sheet where you know what will open it. "Nothing that needs Python" does not mean "opens identically everywhere", and a `#NAME?` in front of a director is a bad way to find out.

**Crux.** XLOOKUP needs Excel 2021 or LibreOffice 24.8 and later; INDEX with MATCH computes everywhere.

3 A lookup must fail visibly

```
=XLOOKUP(id, keys, values, "not in the table")
=IFERROR(INDEX(values,MATCH(id,keys,0)),
  "not in the table")
```

| Written as         | On a missing id        |
|--------------------|------------------------|
| No fourth argument | #N/A                   |
| A not-found string | Your words             |
| Approximate match  | The next larger member |

#N/A is ugly and truthful. A neighbour's row is tidy, wrong and complete-looking, and nobody questions a row that looks complete.

**Crux.** The fourth argument decides whether a missing id returns an error, your words, or a neighbouring member's real data.

5 One number, read correctly

Rs 9.84 crore collected in Q2 across 462 orders, against Rs 10.00 crore in Q1, a fall of 1.6 percent driven by Retail-Plus frequency.

The sentence beside a chart names the branch. A director reading a falling line with no sentence will attach their own reason, and it travels with the chart.

**Crux.** A figure with no denominator, period and comparison gets those three supplied by whoever reads it.

6 The operating rule

| Lives in  | Owns                    | Never                        |
|-----------|-------------------------|------------------------------|
| Warehouse | Numbers Finance acts on | Exploration                  |
| pandas    | The analyst's iteration | The source of truth          |
| Excel     | Presentation and poking | Cleaning, joining, the truth |

**Whatever is in Excel must be rebuildable from the warehouse in one run.** The first hand-typed correction breaks it: either the fix is lost on rebuild, or it is kept and the sheet has become a source of truth nobody chose.

**Crux.** Three homes and one condition, and the condition is the part people forget.